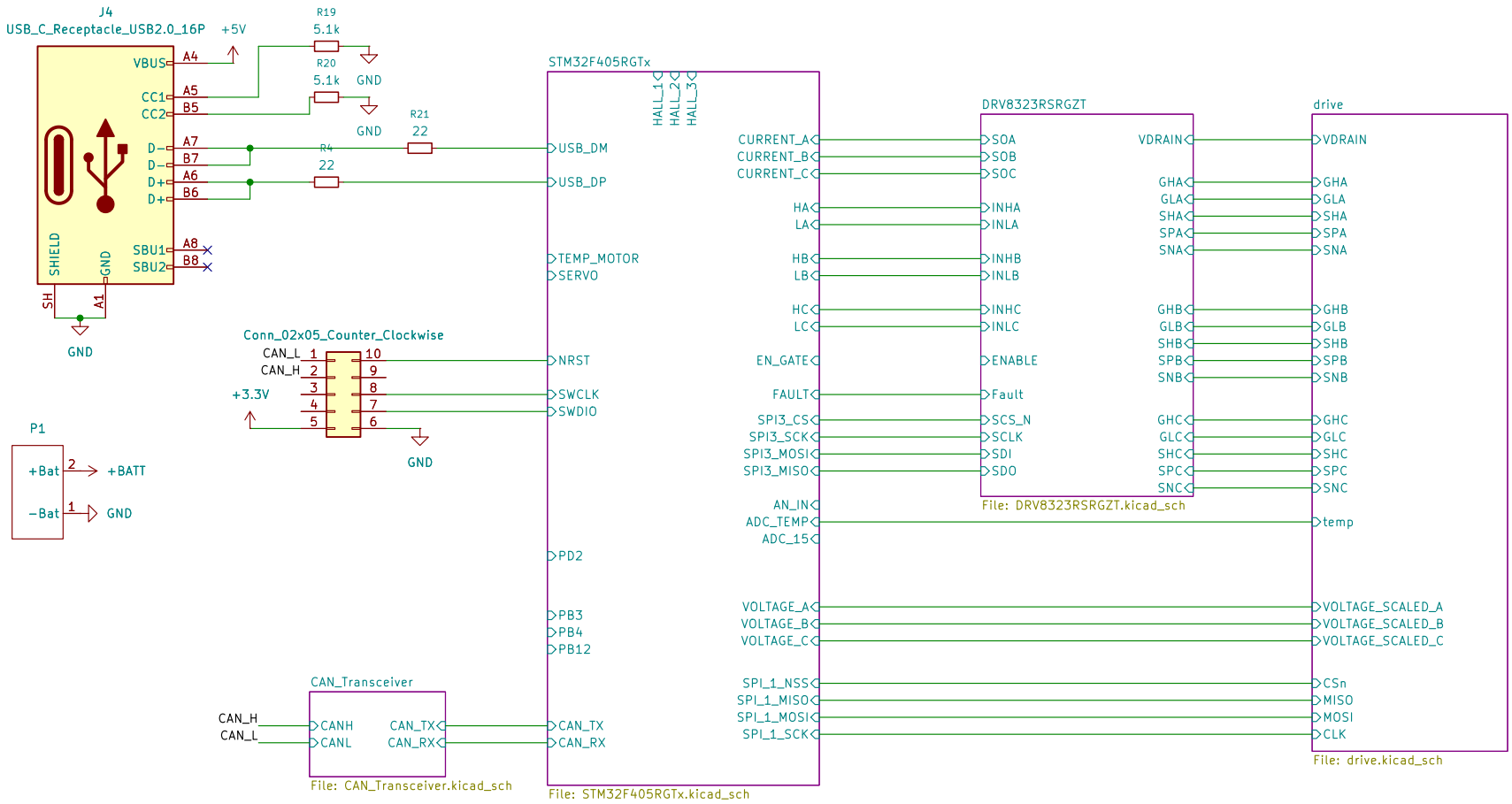
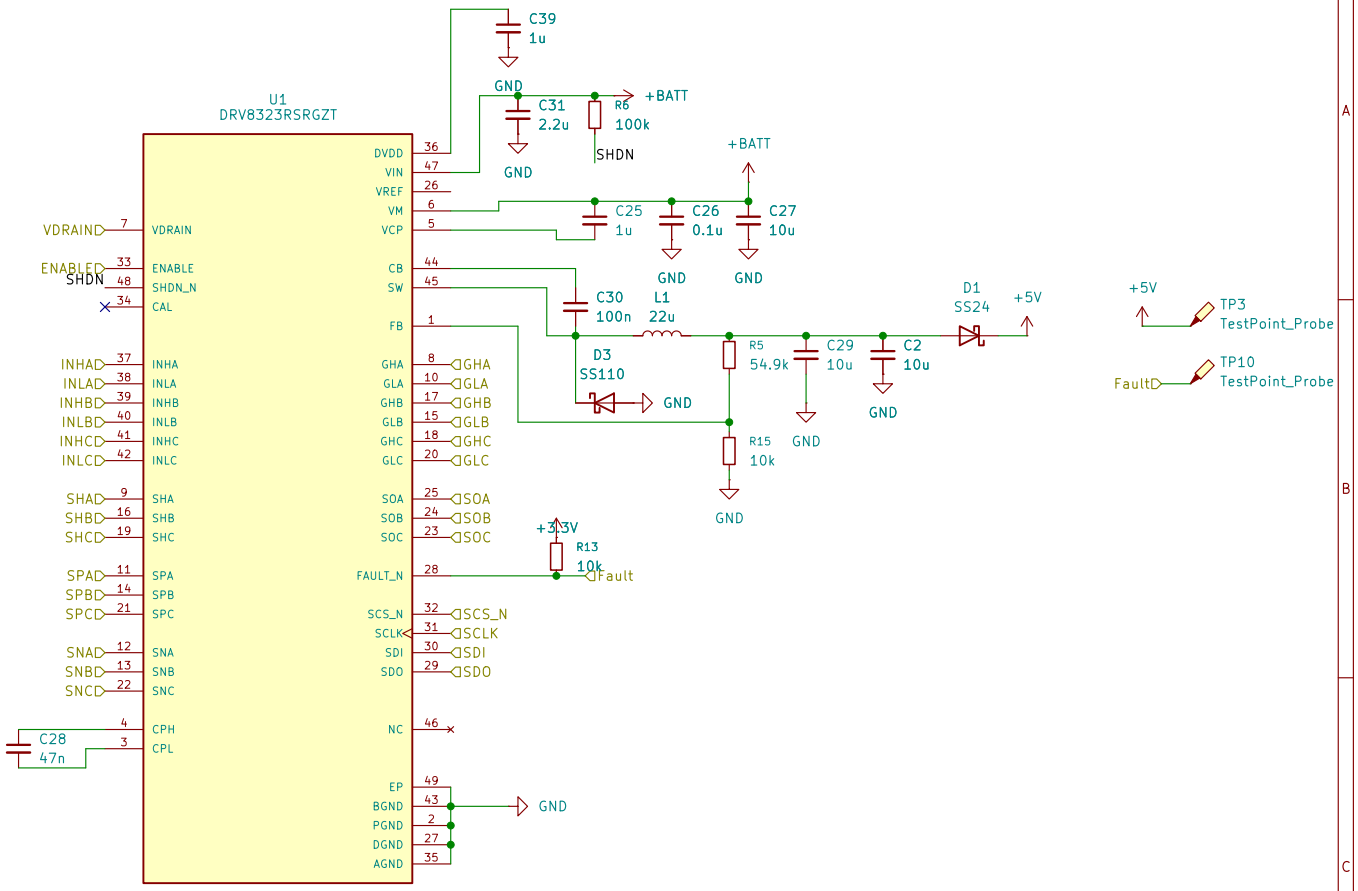
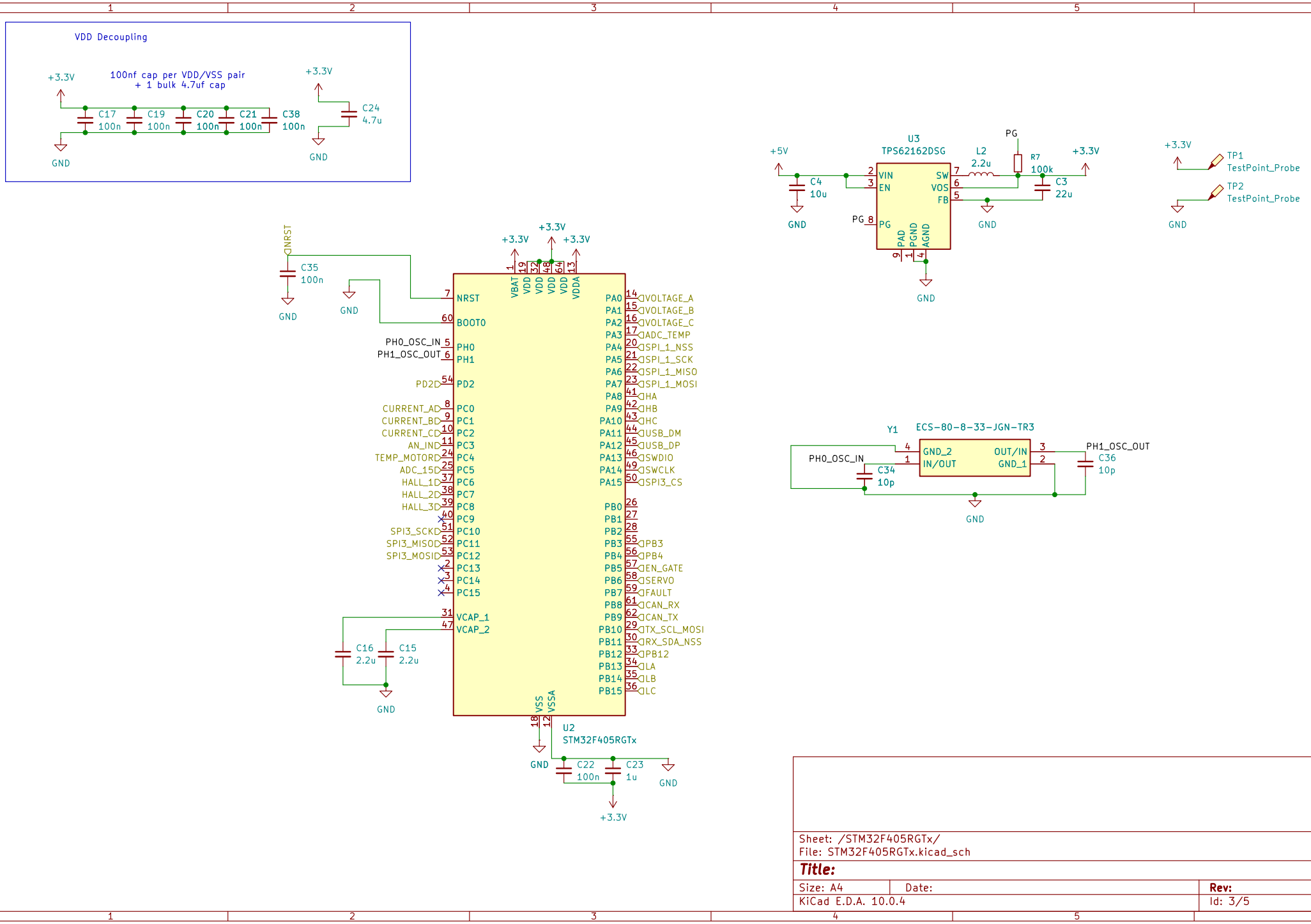
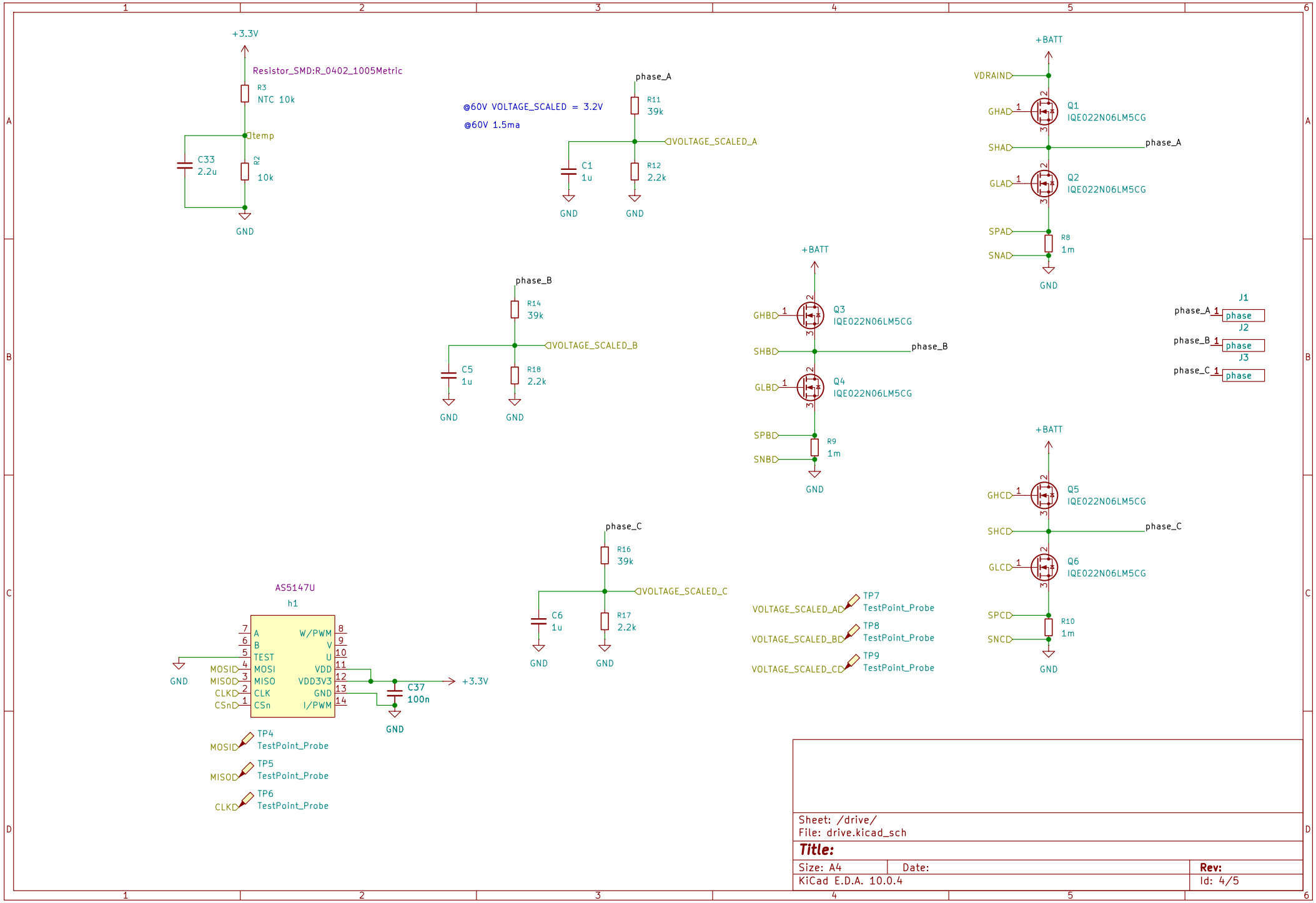


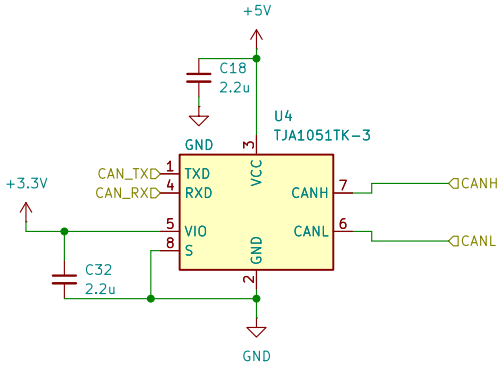
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VDD Decoupling
100nf cap per VDD/VSS pair
+ 1 bulk 4.7uf cap
+3.3V
C17 100n
C19 100n
C20 100n
C21 100n
C38 100n
C24 4.7u
GND
U2
STM32F405RGTx
NRST
BOOT0
PH0_OSC_IN
PH1_OSC_OUT
PD2
CURRENT_AD
CURRENT_BD
CURRENT_CD
AN_IND
TEMP_MOTORD
ADC_15D
HALL_1D
HALL_2D
HALL_3D
SPI3_SCKD
SPI3_MISOD
SPI3_MOSID
VCAP_1
VCAP_2
VBAT
VDD
VDDA
VSS
VSSA
PA0 VOLTAGE_A
PA1 VOLTAGE_B
PA2 VOLTAGE_C
PA3 ADC_TEMP
PA4 SPI_1_NSS
PA5 SPI_1_SCK
PA6 SPI_1_MISO
PA7 SPI_1_MOSI
PA8 HA
PA9 HB
PA10 HC
PA11 USB_DM
PA12 USB_DP
PA13 SWDIO
PA14 SWCLK
PA15 SPI3_CS
PB3 PB3
PB4 PB4
PB5 EN_GATE
PB6 SERVO
PB7 FAULT
PB8 CAN_RX
PB9 CAN_TX
PB10 TX_SCL_MOSI
PB11 RX_SDA_NSS
PB12 PB12
PB13 LA
PB14 LB
PB15 LC
C35 100n
C16 2.2u
C15 2.2u
C22 100n
C23 1u
U3
TPS62162DSG
+5V
L2 2.2u
R7 100k
C4 10u
C3 22u
VIN
EN
SW
VOS
FB
PG
PAD
PGND
AGND
Y1
ECS-80-8-33-JGN-TR3
C34 10p
C36 10p
GND_2
IN/OUT
OUT/IN
GND_1
TP1 TestPoint_Probe
TP2 TestPoint_Probe
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+3.3V
Resistor_SMD:R_0402_1005Metric
R3
NTC 10k
temp
C33
2.2u
R2
10k
GND
@60V VOLTAGE_SCALED = 3.2V
@60V 1.5ma
phase_A
R11
39k
VOLTAGE_SCALED_A
C1
1u
R12
2.2k
GND
GND
+BATT
VDRAIN
GHAD
Q1
IQE022N06LM5CG
SHAD
phase_A
GLAD
Q2
IQE022N06LM5CG
SPAD
SNAD
R8
1m
GND
phase_B
R14
39k
VOLTAGE_SCALED_B
C5
1u
R18
2.2k
GND
GND
+BATT
GHBD
Q3
IQE022N06LM5CG
SHBD
phase_B
GLBD
Q4
IQE022N06LM5CG
SPBD
SNBD
R9
1m
GND
J1
phase_A
phase
J2
phase_B
phase
J3
phase_C
phase
+BATT
GHCD
Q5
IQE022N06LM5CG
SHCD
phase_C
GLCD
Q6
IQE022N06LM5CG
SPCD
SNCD
R10
1m
GND
AS5147U
h1
A
B
TEST
MOSI
MISO
CLK
CSn
W/PWM
V
U
VDD
VDD3V3
GND
I/PWM
GND
MOSID
MISOD
CLKD
CSnD
C37
100n
+3.3V
GND
phase_C
R16
39k
VOLTAGE_SCALED_C
C6
1u
R17
2.2k
GND
GND
VOLTAGE_SCALED_AD
VOLTAGE_SCALED_BD
VOLTAGE_SCALED_CD
TP7
TestPoint_Probe
TP8
TestPoint_Probe
TP9
TestPoint_Probe
MOSID
MISOD
CLKD
TP4
TestPoint_Probe
TP5
TestPoint_Probe
TP6
TestPoint_Probe
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